

AARYA DHALIWAL

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Education

University of Houston

Bachelor of Science in Computer Science, GPA: 3.3

Aug. 2023 – May 2027

Houston, Texas

Relevant Coursework

- Intro To Programming
- Computer Architecture
- Data Structures
- Database Management

Experience

Best Brains

December 2024- April 2025

Coding/Abacus Instructor

Cypress, Texas

- Taught Coding such as Scratch, Python, HTML, and Abacus to students aged 5–14, fostering a love for learning and critical thinking skills.
- Designed and delivered engaging, interactive lessons tailored to diverse student needs and skill levels.
- Monitored student progress, providing feedback and support to help them achieve their learning goals.
- Collaborated with management and parents to ensure a cohesive educational experience.
- Developed strong communication and classroom management skills in a dynamic learning environment.

iCode

Dec 2023 – March 2024

Coding Instructor

Katy, Texas

- Developed effective lesson plans to keep students aged 6-15 engaged and accommodated.
- Met the educational needs of students through visual, auditory, and hands-on learning methods.
- Provided individualized support to students through after-school tutoring and office hours.
- Utilized a variety of technology tools to engage students and promote critical thinking skills in coding and non-coding environments
- Analyzed student data to identify areas of weakness and adjust instruction to better meet student needs.

Projects

University Database Web Application | JavaScript HTML/CSS, Postman (COSC3380 Coursework)

October 2025

- Developed a full-stack web application for managing university data.
- Implemented RESTful APIs using Node.js and Express to handle CRUD operations on a PostgreSQL database storing student and faculty information.
- Designed and styled the frontend interface with HTML/CSS for data input and display.
- Tested endpoints and validated responses using Postman to ensure accurate data retrieval and integration between frontend and backend systems.

Smart Playlist Manager with Favorites | C++ (COSC2436 Coursework)

February 2025

- Designed and implemented a music playlist manager using a custom **doubly linked list** data structure
- Developed a **smart playlist generation algorithm** that optimally selects songs under a time limit, prioritizing favorite songs before filling remaining time with non-favorites
- Strengthened problem-solving skills in memory management, pointer manipulation, and algorithm design

Casino Card Game Simulator | C++ (COSC2436 Coursework)

May 2025

- Implemented a queue-stack simulation to model a casino card game where participants request specific numbers of red and black cards
- Designed data structures to manage participant order (queue) and dealer card distribution (stack)
- Developed robust input/output parsing for participant credentials and dealer card stack with error handling for empty cases (“No Game!”, “No Loser!”)

Technical Skills

Spoken Languages: Hindi

Languages: Python, Java, C++, HTML/CSS, JavaScript, SQL

Developer Tools: VS Code, Git, GitHub, Google Cloud Platform, PyCharm, IntelliJ IDEA, MySQL Workbench, pgAdmin4, Postman, PostgreSQL

Technologies/Frameworks: Linux, GitHub, React, Windows, Node.js, Express

Extracurricular

CougarCS

Spring 2025 – Present

Member

University of Houston

- Attended career advancement events that included mock interviews and resume building workshops.
- Worked as a team in a social setting to solve problems together.
- Explored connections and collaborated with professionals during club-sponsored events.